Unit 3 – Respiration and Excretion

Content :

* Respiratory system.
* Gaseous exchange.
* Respiration in other organisms.
* Anaerobic respiration.
* Excretion – Structure of Kidney.
* Formation of urine.
* Kidney health care.
* Excretion in other organisms.

Unit summary :

Respiration is the process of releasing energy from food. While aerobic respiration takes place using oxygen, anaerobic respiration does not. Ventilation (breathing) takes place to deliver oxygen to the human body. The alveoli are delicate air sacs in the lungs that serve as the respiratory surface for receiving oxygen. Diffusion of oxygen occurs into the blood vessels that cover the alveoli. Along with this, carbon dioxide also diffuses into the alveoli.

Most of the oxygen that enters the blood travels with hemoglobin as *oxyhemoglobin*, liberating oxygen into the tissue fluid. This oxygen reaches the mitochondria that make energy inside the cells, and with the help of enzymes, the energy in the food is released. This activity that takes place in the cells is known as cellular respiration. After *glycolysis*, the second phase, (the *Krebs cycle)* occurs in the mitochondria.

Cell membrane in amoeba, skin in earthworm and gills in fish serve as their respiratory surface. In plants, diffusion of respiratory gases occurs through stomata, lenticels and cells of root.

Through various metabolic activities in the cells, harmful substances to internal environment can also be produced. Excretion is the process of expelling these wastes. *Urea*, which is formed from ammonia, a toxic substance produced as a result of protein metabolism, is one of the main excretory substances in our body. A pair of kidneys are present in our body to filter the urea produced in the liver from the blood and excrete it through the urine.

About 12 lakh nephrons in each kidney filter urea from the blood. When blood passes through the glomerulus (minute capillaries present in the Bowman's capsule of the nephron), the blood undergoes ultrafiltration and a liquid called the glomerular filtrate is formed. From this fluid, the essential components needed by the body are reabsorbed into the blood. What is left is urine. It reaches to the bladder through the ureters.

Kidney stones, nephritis, uremia etc are disorders affecting the kidneys. When both kidneys fail, hemodialysis and kidney transplant are used to purify the blood and maintain life.

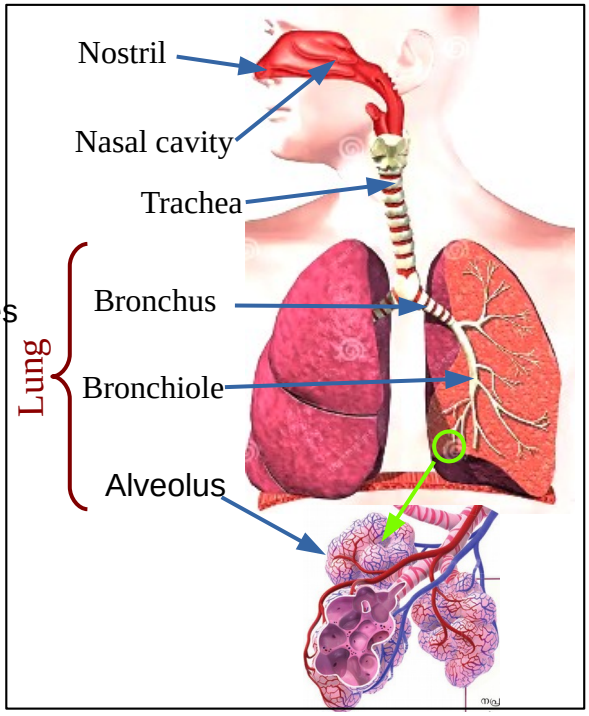
Different species of animals have kidneys or other organs to excrete waste such as ammonia, urea and uric acid. Contractile vacuole in amoeba, nephridia in earthworms, and Malpighian tubules in insects perform the function of the kidney. In plants, excretory materials are released through stomata, lenticels, hydathodes, mature leaves and other means.

Homeostasis of the body is maintained by the kidneys, liver, lungs, and skin through eliminating waste materials. But pollution, lack of hygiene, alcohol, smoking, drugs, unhealthy food habits, mental stress, etc. can disrupt the internal environment and lead to ill health.

Questions & Answers :

1. Respiration ?
   Respiration is the process of releasing energy from food.
   Animals and plants use oxygen for respiration.

2. The organ that acts as a respiratory surface to receive oxygen in humans?
   Lungs.
3. The elastic membraneous sacs within the lungs where gas exchange takes place?
   Alveoli.
4. Parts of human respiratory system ?
   (Pathway by which atmospheric air reaches the alveoli)
   Nostrils – Nasal cavity – Trachea – Bronchi – Broncheoles – Alveoli.
5. What causes increased respiratory surface area?
   About 70 crore alveoli found in both lungs increase the respiratory surface area.
6. The membrane which covers the lungs?
   Pleura.
7. The muscular wall which separates the thoracic and abdominal cavities. ?
   Diaphragm.
8. The muscles seen in between the ribs?
   Intercostal muscles.



9. The stages in our respiratory process?
   Ventilation (breathing), Exchange of gases and Cellular respiration.

10. What is ventilation?
    Movement of air from the atmosphere to lungs and back is called as ventilation.
    [Include the entering of air or inspiration (inhalation) and going out of air or expiration (exhalation)]

11. How does ventilation occur in humans?
    By the contraction of the intercostal muscles,the ribs raise. And the diaphragm contracts. Hence, the volume of thoracic cavity increases and air pressure inside the lungs reduces. Then the air enter inside the lungs. This is called inspiration (inhalation).
    When the intercostal muscles and diaphragm returns to its original condition, the volume of thoracic cavity decreases and air pressure inside the lungs increases to go the air outside. This is called expiration (exhalation).

12. Factors that help alveolar exchange of gases ?
    The wall of alveolus and wall of blood cpillary are made of single layer of cells. Moisture inside alveolus helps to dissolve gases. A difference in the concentration of gases also makes diffusion easy. Both the CO_2 and O_2 can easily attach with haemoglobin present in the RBCs.

13. The process of gaseous exchange between blood and alveoli?
    Most of the oxygen entering the blood combines with hemoglobin in red blood cells to form oxyhemoglobin. Some oxygen dissolves in the plasma (Alveolar exchange of gases).
    Oxygen traveling through the blood, diffuses into the tissue fluid and then into the cells. The CO_2 diffuses back into the tissue fluid and then into the blood (Systemic exchange of gases).

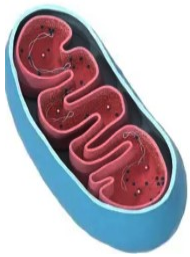
    70% CO_2 dissolved in red blood cells as bicarbonate, 23% of CO_2 combines with haemoglobin as carbaminohemoglobin and 7% CO_2 dissolved in plasma. The CO_2 released by the breakdown of carbaminohemoglobin and bicarbonate diffuses from the blood into the alveoli.

14. O_2 + Haemoglobin = Oxyhaemoglobin,
    CO_2 + Haemoglobin = ---------- ?
    Carbaminohaemoglobin.

15. A way to understand that exhaled air contains more CO_2 than inhaled air?
    * When we blow exhaled air into clear lime water, it turns a milky color.
    * When we blow exhaled air into red colored hydrogen carbonate solution, it turns yellow.

16. Cellular respiration?
Cellular respiration is the process of breaking down glucose with oxygen inside the cell to release energy.
It involves two phases:- A. Glycolysis B. Krebs cycle.

17. Compare between the phases glycolysis and Krebs cycle.
Glycolysis -The process by which glucose is converted to pyruvic acid occur with in the cytoplasm.
It does not require oxygen. 2 ATP energy molecules are formed.
Krebs Cycle - The process occur with in the mitochondria by which pyruvic acid is converted to CO_2 and water with the help of oxygen and enzymes.
28 ATP molecules are formed.


Mitochondrion

Glucose + Oxygen $\xrightarrow{\text{Enzymes}}$ O_2 + Water + 30 ATP

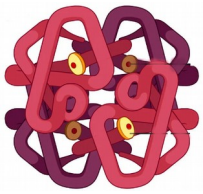
18. Why is the second phase of cellular rspiration called as Krebs cycle?
It was Hans Adolf Krebs, a German biochemist, discovered the chemical processes in the second phase of cellular respiration.



19. Comparison of the metabolic processes photosynthesis and respiration:

Photosynthesis	Respiration
- Anabolism (synthesis). - Light energy is assimilated as chemical energy with in the food molecules. - CO_2 and water are used. - O_2 is expelled out. - Light phase occur in the grana and the dark phase in the stroma.	- Catabolism (breakdown). - The process of releasing energy from food. - CO_2 and water are expelled. - O_2 is used. - Glycolysis occur in the cytoplasm and the Krebs cycle in the mitochondria.

20. The pigment in the RBCs for transporting oxygen and CO_2?
Haemoglobin. (Iron contained protein is the main componet)
Each haemoglobin can carry **four** molecules of oxygen or CO_2.



21. The normal level of haemoglobin?
In females, 12-16 gm/dL blood. In mails,14-18 gm/dL blood.

22. The condition in which the level of haemoglobin decreases in blood ? Reason?
Anaemia.
Deficiency of iron is the main reason. Other reasons include deficiency of certain vitamins, certain diseases, genetic disorders.

23. Mode of respiratory gaseous exchange in amoeba, earthworm and fish?

	Respiratory surface	Exchange of gases
Amoeba	Cell membrane	Between external environment and cytoplasm.
Earthworm	Skin	Between atmospheric air and blood in the cpillaries of skin.
Fish	Gill	Between aquatic air and blood in the cpillaries of gills.

24. Mechanisms of gaseous exchange in plants?
Stomata, Lenticels (small pores on the surface of mature stem and roots) and Cells of root.

25. Changes in gas exchange in plants during day and night ?
During the day, most of the CO_2 for photosynthesis enters through the stomata. O_2 released during photosynthesis is also used by plants for daytime respiration.
During night when stomata close,the exchange of respiratory gases takes place through diffusion.

26. Differentiate between aerobic and anaerobic respiration ?
Aerobic respiration is the process of respiration using oxygen. Anaerobic respiration takes place without the use of oxygen. (Bacteria and yeast do not need oxygen for energy production)

Anaerobic respiration involves only glycolysis. So, only 2 ATP molecules are obtained.

27. Compare the anaerobic respiration process in Lactobacillus bacteria and yeast.

In Lactobacillus bacteria, glucose is converted to pyruvic acid and then to lactic acid.
(eg:- Milk turns into curd).

In Yeast, glucose is converted to pyruvic acid and then to alcohol and CO_2
(eg:- Yeasted dough will leaven and become soft).

28. An instance of anaerobic respiration in humans ?

When certain muscles are continuously worked, the oxygen in the muscle cells decreases and anaerobic respiration takes place to form lactic acid there. This causes muscle pain.

29. What is excretion??

Excretion is the process of removing harmful waste materials from the body.

30. Although CO_2 is an excretory product, it is beneficial to the body. How?

CO_2 is needed to convert ammonia, a toxic substance produced in cells as a result of protein metabolism, into an excretory substance called urea, in the liver.

31. How is urea formed?

As a result of protein metabolism in cells, ammonia, a toxic waste is formed and reaches in blood. When it reaches the liver, it reacts with CO_2 and water with the help of enzymes to form a less toxic urea.

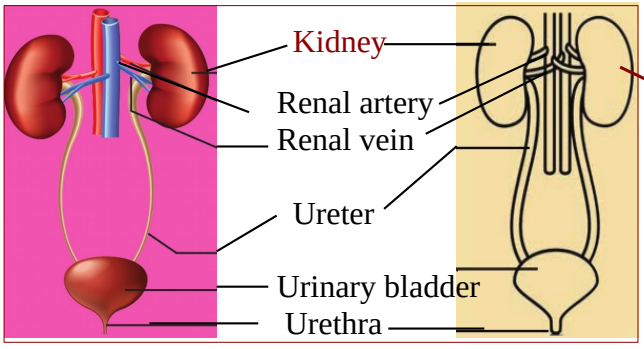
Protein metabolism → **Ammonia + CO_2 + Water** → **Urea**

Both the kidneys filter waste materials like urea from blood and remove from the body.

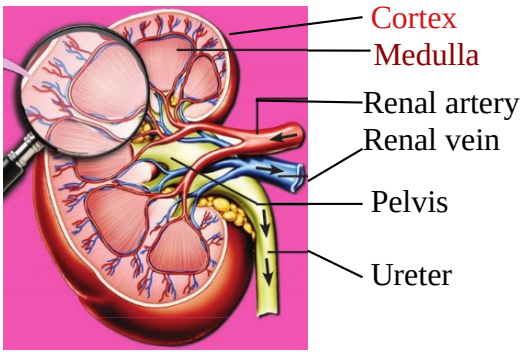
32. The excretory organs in man?

Kidneys - (water and salts are excreted through urine), **Liver** – (synthesize urea),
Lungs – (expel CO_2 through exhaled air),
Skin - (water and salts are excreted through sweat).

33. Kidneys and related parts:



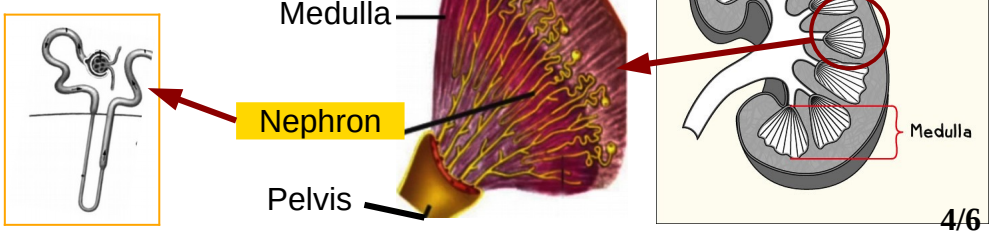
Structure of a kidney:



In humans, a pair of pea-shaped kidneys are found in the abdomen, on either sides of vertebral column. The renal artery is the vessel that carries blood to the kidneys. Renal vein carries blood away from the kidneys. Ureter carry urine from kidney to the urinary bladder, from which urine is expelled via urethra.

About 12 lakh nephrons in each kidney helps in the formation of urine by filtering urea and other waste material from the blood. Cortex, the light coloured outer part of the kidney contains minute capillary network of the nephron. The dark coloured inner part of kidney is medulla. Pelvis is the initial part of ureter.

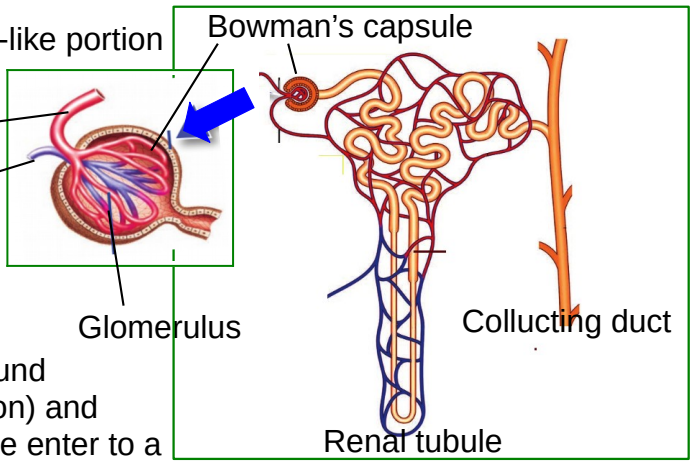
34. The basic functional units in kidneys?
(The ultra filters)?
Nephrons.



35. Structure of nephron:

A long, coiled nephron has a double-walled cup-like portion at one end. This is called Bowman's capsule. A small branch of the renal artery supplies blood to it. This is afferent vessel, which turned into a network of minute capillaries, called the glomerulus. (ultrafiltration takes place through the pores of glomerulus).

An efferent vessel is a small diameter tube that carries blood away from the glomerulus. This inturn becomes another network of vessels around the long renal tubule of nephron (for reabsorption) and extended as a branch of renal vein. Renal tubule enter to a collecting duct, collect urine and carry it into the pelvis of ureter.



36. How urine is formed? (stages)

When blood passes through the glomerulus, seen in the Bowman's capsule of the nephron, it undergoes ultrafiltration and produces a liquid called glomerular filtrate. From this, the essential substances are reabsorbed into the blood. What is left is the urine.

1. **Ultrafiltration** –_ Blood is filtered through the pores of glomerulus and glomerular filtrate is formed. Besides the waste materials, it contains essential substances too.
2. **Reabsorption** – When glomerular filtrate flows through renal tubules to the collecting duct, the essential components are reabsorbed from it into the blood.
3. **Secretion** – The excess components that are retained in blood are secreted into the renal tubules. The remaining fluid flows to the pelvis through the collecting duct is called urine. It reaches to the bladder through the ureters.

37. Flowchart showing the formation and excretion of urine:

(blood) -- Aorta – Renal artery -- Glomerulus -- (glomerular filtrate)--Renal tubule –Collecting duct -- (urine) -- Pelvis-- Ureter – Urinary bladder – flows out through Urethra.

ultrafiltration *Reabsorption, secretion*

38. Compare the components of glomerular filtrate with urine and find out which are reabsorbed into the blood.

Components of glomerular filtrate	Components of urine
Water, glucose, amino acids, vitamins, Sodium, potassium and calcium ions, Urea, uric acid, creatinine etc.	Water, urea, sodium chloride, potassium chloride, calcium ions,phosphate, uric acid, creatinine etc.

Glucose, amino acids, vitamins and a little amount of water are reabsorbed to the blood.

39. Every person should drink plenty of clean water and urinate frequently. Give reason ?

The washing out of disease causing germs from the urinary tract takes place during the process of micturition. Otherwise, infection of inner membrane occur leading to kidney diseases.

40. Examples of diseases that can be detected by urine test?

Diabetes (presence of glucose),
Jaundice (presence of bilirubin),
Kidney diseases (presence of blood or albumin),
Urinary tract infections (pus cells),
Kidney stone (calcium oxalate crystals.)

41. A test to determine the presence of glucose in urine:

Take 2ml of urine in a test tube and add 2ml of Benedict's reagent to it.. After heating for two minutes observe the color change.

(If it is blue, there is no glucose. If it is green, less glucose, ie, below 1%. If it is yellow, 1.5% glucose. Orange colour is dangerously high up to 2%. Red shows more than 2% is very dangerous)

42. Diseases that affect the health of kidney?
Kidney stone - (abdominal pain, blockage of urine, dizziness, vomiting due to the deposition of calcium crystals).
Nephritis - (infection or intoxication cause inflammation, turbid urine and oedema).
Uremia –(non functioning of kidneys by nephitis, diabetes etc, Stops urine formation)

43. A way to keep one's life alive if both kidneys stop working?
Purification of blood via hemodialysis, Kidney transplantation.

44. Steps in the process of hemodialysis:
- Blood is passed tothe dialysis unit after adding heparin to prevent coagulation.
- When blood flows through the tube made up of selectively permeablemembrane, the waste materials diffuse into the dialysis fluid. This fluid is removed in due course.
- Anti heparin is added to the purified blood and is returned to the body.

45. When is kidney transplantation required?
When both the kidneys are not functioning.

46. Things to watch out for during kidney transplant surgery?
A kidney is transplanted from a healthy or a healthy person who has died in accident or by other means. Blood group matching, tissue matching and cross matching will be done prior to the transplant. Being another person's kidney, medicines which suppress immunity are to be used.

47. Excretory mechanisms in amoeba and other organisms:

	Excretory product	Excretory mechanism
Amoeba	Ammonia	Contractile vacuole
Earthworm	Ammonia, Urea	Nephridia
Insect	Uric acid	Malpighian tubule
Fish	Ammonia	Kidneys
Frog	Urea	Kidneys
Reptile, Bird	Uric acid	Kidneys

48. Excretory mechanisms in plants ?
CO_2 and O_2 are excreted through the stomata and lenticels.
Water and salts are excreted through hydathodes.
Other solid wastes are eliminated by resins, bark, dropping of mature leaves and fruits, formation of heart wood etc.

49. How the human excretory system contributes to homeostasis?
Homeostasis of the body is maintained by the kidneys, liver, lungs, and skin through eliminating waste materials.
Skin – regulates body temperature, salts and water.
Liver – regulates metabolism and neutralises toxins.
Kidneys – besides expelling wastes, salt-water balance, regulation of pH and BP.
Lungs – elimination of CO_2 and regulation of O_2 as well as the pH.

50. Things that disturb the homeostasis of body?
Unhealthy food habits, over nutrition, under nutrition, lack of excercise, mental stress, alcohol, smoking, additive drugs, pollution, lack of hygiene, pathogens etc. Can disrupt the internal environment and lead to ill health.